

BENJAMIN E. JORDAN

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LinkedIn | Personal Website | GitHub

EXPERIENCE

Software Engineer @ Amazon

Oct 2024 – Present

Measurement, Ad Tech, & Data Science

New York, NY

- Owned Brand Lift team's production ML system (built on AWS SageMaker), which processes billions of rows of data daily, producing modeled lift estimates and aggregating survey responses for reports
- Independently designed and implemented a daily feature ingestion job + data store, reducing batch inference costs by over \$100k per year (roughly 50% cost reduction) and enabling the system to scale cheaply to new marketplaces/regions
- Designed and implemented a workflow to automate model training, evaluation, and approval on a weekly cadence
- Worked with scientists to enable shadowing of experimental models on production data
- Presented system to teammates and engineers from sister teams within the org
- Led the design and development of a reporting feature which automatically compiles study results into presentation-ready PowerPoints, saving hours of manual effort for thousands of users with active studies

Machine Learning Engineer @ Northrop Grumman

Feb 2024 – Oct 2024

Autonomous Intelligence & Robotics Lab

Denver, CO

- Enabled accurate, real-time object detection by distilling GroundingDINO predictions into YOLOv10, decreasing inference latency on Jetson Orin from 0.6s to 0.08s
- Researched and evaluated techniques for dealing with a lack of labeled data; active learning, semi-supervised learning, knowledge distillation, and synthetic data generation, informing the team's adoption of distillation as a tool for efficiency
- Led the technical development of a project focused on classified PDF retrieval and question answering, and wrote a report covering techniques for implementing and evaluating retrieval-augmented generation (RAG) systems
- Implemented a RAG system utilizing ANN search and locally running language and dense retrieval models, evaluated retrieval performance on dataset of LLM generated Q&A pairs based on documents using MRR and Recall@K

Machine Learning Engineer @ KLA

May 2023 – Aug 2023

Internship

Ann Arbor, MI

- Researched and implemented algorithms for semiconductor defect detection and classification
- Fine-tuned and quantized vision transformers with limited data, outperforming statistical methods while staying efficient
- Presented project during poster board session and was invited to give a virtual talk on transformers and transfer learning

Software Engineer @ Carestream

May 2022 – Aug 2022

Internship

Rochester, NY

- Software engineering internship working on fullstack software for x-ray machines

PROJECTS

Entropy Audio | Website | Project Blog

Jan 2024 – Present

Personal project focused on novel sound design tools for composers

- Trained a 1.4B diffusion transformer on a multi-terabyte audio dataset, adding text conditioning from T5Gemma, CLAP/BPM/key global conditioning, and measured performance with a wide set of offline evaluation metrics
- Used local models (Gemma 4 31B), an RTX 5090, and vLLM to efficiently generate captions for millions of audio samples (3TB), and built pipelines for embedding-based deduplication, quality filtering, and dataset balancing
- Implemented preference/post-training methods for diffusion models, including Flow-GRPO policy-gradient optimization on pretrained audio quality models, and Diffusion-DPO on a self-collected preference dataset
- Built an audio-generation service using Lambda, S3, DynamoDB, and RunPod GPU endpoints, orchestrating model inference, generation metadata and audio storage, and preference data collection
- Designed a website that incorporates a preference data flywheel into the generation workflow, where user selections from a batch are treated as positive/preferred and other items in the batch are treated as negatives

Open Source Contributions | GitHub

April 2026 – Present

- Minor contributions to Google's JAX, a library for high-performance numerical computing and large-scale machine learning

SKILLS

Languages: Python, Scala, C++ , TypeScript, Java **Infra:** Docker, AWS, SageMaker, EMR, Linux, Runpod

Libraries: PyTorch, JAX, Spark, NumPy, Pandas, Weights & Biases, vLLM, Faiss, Optuna, OmegaConf, HuggingFace

EDUCATION

Cornell University

2022 – 2023

Master's in Computer Science, Concentration in Machine Learning

Ithaca, NY

Rochester Institute of Technology

2018 – 2022

Bachelor's in Computer Science, Minor in Music, Magna Cum Laude, Presidential Merit Scholarship

Rochester, NY